

SRIKRUTHI KUDUMULA

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PROFESSIONAL SUMMARY

Results-driven **Software Engineer** with **5+ years** designing and deploying **enterprise-scale web applications** across **clinical research, insurance, and eCommerce domains**. Full-stack expertise in **C#, ASP.NET/Web APIs, Angular, React, SQL Server, and cloud platforms (Azure, AWS)**. Experienced building scalable **RESTful APIs, microservices, and automated data migration tools** serving thousands of users and processing millions of transactions daily. Strong track record implementing **CI/CD pipelines, containerization (Docker/Kubernetes), and observability solutions (Application Insights, CloudWatch)** to enhance reliability and optimize performance in Agile environments. Proven success delivering FDA-compliant clinical systems for 50+ trial sites, insurance licensing platforms for 50K+ daily users, and high-volume eCommerce integrations processing 80K+ daily requests with consistent reliability. Skilled in **database optimization, SSIS ETL pipelines, and large-scale Oracle to SQL Server migrations**.

PROFESSIONAL EXPERIENCE

Client: Medpace, Inc.

Role: Software Engineer

Duration: Mar 2023 – Oct 2025

Key Technologies: Angular, React, C#, Git, MS SQL Server, Azure DevOps, Azure App Services, Power BI, SharePoint, WinForms, ASP.NET/Web APIs, TFS

- Architected and deployed 5+ scalable web applications using **Angular, React, and C# on Azure App Services**, implementing auto-scaling policies and load balancing to ensure high availability for clinical trial operations.
- Developed CRO Logistics inventory management web application using **ASP.NET Web APIs and Angular**, enabling real-time equipment tracking and resource allocation across multiple clinical trial sites.
- Created automated SharePoint-to-TMF migration tools using **C#, ASP.NET, and WinForms** to transfer trial documentation while preserving folder hierarchies and metadata integrity.
- Developed **SQL Server stored procedures and data validation scripts** for SharePoint-to-TMF migrations, eliminating 500+ hours of manual data entry annually and improving data accuracy.
- Implemented **SQL Server indexing strategies and query optimization techniques** to improve data retrieval performance for clinical trial information systems.
- Integrated **Power BI** dashboards into CRO Logistics platform to visualize inventory metrics, equipment utilization rates, and trial site distribution analytics for clinical operations teams.
- Built secure **clinical data capture endpoints** with FDA-compliant audit trails, ensuring regulatory compliance and electronic data collection integrity across 50+ trial sites with real-time synchronization.
- Optimized **Azure App Services** performance by configuring auto-scaling rules based on CPU and memory thresholds, ensuring consistent application responsiveness during peak clinical trial activities.
- Integrated **Application Insights monitoring and Azure Monitor alerting** across clinical applications to track performance metrics, detect anomalies, and ensure rapid incident response for trial-critical workflows.
- Designed and maintained CI/CD pipelines in **Azure DevOps** with automated build processes, unit testing gates, and multi-environment deployments, reducing deployment time by 60% and enabling faster feature delivery.
- Implemented **Git branching strategies** with feature branches, pull requests, and code review standards using **TFS** and **Git** for version control, improving code quality and facilitating coordination across distributed development teams in 2-week Agile sprints.

Client: Vertafore

Role: Software Engineer

Duration: Nov 2020 – Nov 2021

Key Technologies: ASP.NET/Web APIs, C#, MS SQL Server, NUnit, SonarQube, Microsoft Azure, Angular, Bootstrap, Entity Framework, Docker, Kubernetes, LINQ

- Engineered Sircon for Carriers insurance licensing platform using **Angular and Bootstrap frontend with ASP.NET Web APIs, Entity Framework, and LINQ backend**, enabling real-time license verification and compliance tracking for 50K+ daily users with improved transaction throughput.

- Developed **SQL Server stored procedures and optimized database queries** to enhance license verification workflows, reducing response times for compliance tracking operations.
- Streamlined **CI/CD pipelines using Azure DevOps** to automate build, test, and deployment processes, enabling faster and more reliable releases with minimal downtime.
- Managed **Azure cloud infrastructure** including Virtual Machines, Blob Storage, Azure SQL databases, and Azure Active Directory to ensure scalable and highly available deployments.
- Configured and maintained **Docker containerized applications, integrating with Azure Kubernetes Service (AKS) clusters** to ensure consistent deployments and reliable application availability across development, staging, and production environments.
- Implemented **NUnit-based** unit testing frameworks across service layers, achieving 80% test coverage to enhance code quality and reduce production defects.
- Improved code maintainability and reduced technical debt through **SonarQube code analysis**, addressing code smells, security vulnerabilities, and complexity issues.
- Collaborated with cross-functional and offshore teams in Agile sprints to deliver customized enhancements and resolve production issues efficiently

Client: ShopClues

Role: Application Developer (Part-time)

Duration: Jan 2019 – Jul 2020

Key Technologies: C#, ASP.NET/Web APIs, MS SQL Server, Oracle, SSIS, ADO.NET, React, Bootstrap, AWS (EC2, CloudWatch, X-Ray)

- Integrated payment gateway services using **ASP.NET Web APIs, ADO.NET, and MS SQL Server** to process online transactions, maintaining 99.7% uptime while handling 80K+ daily payment requests.
- Designed responsive seller management portal using **ASP.NET Core, React, HTML/CSS, and Bootstrap, hosted on AWS EC2 instances** to enable 100K+ sellers to manage listings, inventory, and marketing campaigns with faster onboarding.
- Built logistics partner integration APIs using **ASP.NET Web APIs** to connect third-party shipping providers with the eCommerce platform, enabling automated order fulfillment workflows across 500K+ product listings.
- Developed order processing system backend using **ASP.NET and ADO.NET** to handle order workflows from checkout to fulfillment, improving operational efficiency and reducing manual processing time.
- Built order processing frontend using **React and Bootstrap** to provide real-time order status updates, streamlining user experience for customers and internal operations teams.
- Led database migration from **Oracle to SQL Server 2016**, restructuring monolithic database architecture into modular, service-aligned databases to improve query performance and scalability.
- Created **multithreaded catalog migration tools using C#** to transfer 5M+ SKUs from legacy systems with minimal downtime, implementing data integrity validation and error handling mechanisms.
- Developed **SSIS ETL** pipelines to automate product catalog synchronization, ensuring real-time inventory updates across marketplace and seller systems with scheduled batch processing.
- Configured **AWS CloudWatch and X-Ray** for ShopClues applications to track performance metrics, set up custom alerts, and detect issues proactively to improve system reliability.
- Collaborated with product and infrastructure teams in Agile sprints to optimize marketplace scalability, implement new seller features, and ensure smooth feature rollouts across platforms.

TECHNICAL SKILLS

Programming Languages: C#, Java, Python, JavaScript, TypeScript

Web Technologies: HTML5, CSS3, Bootstrap, WinForms, ASP.NET(MVC, Webforms), RESTful APIs, ADO.NET

Frameworks & Libraries: Angular, .NET/.NET Core, Web API | React, LINQ, Entity Framework

Databases: MS SQL Server (T-SQL), Oracle (PL/SQL), SQLite

Cloud/DevOps: Azure (App Services, DevOps), AWS, CI/CD, Docker, Kubernetes

Tools & Platforms: Visual Studio, VS Code, Power BI, SSIS, SharePoint, Jira, SonarQube, Git, TFS

Testing & Methodologies: NUnit, Unit Testing, TDD | Agile, Scrum, Waterfall, SDLC, OOP, Microservices Architecture

EDUCATION

Master of Science, Information Technology — University of Cincinnati | Jan 2022 – Apr 2023 | GPA: 3.9/4.0

Bachelor of Technology, Computer Science and Engineering — JNTU Hyderabad | Aug 2016 – Nov 2020 | GPA: 3.0/4.0